Urban service allocation: traffic and sensitivity specification

Urban traffic methods

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Purpose and scope. This document specifies a discrete continuous-mass model for an urban signal-control task. It defines lane admission, order-dependent access, service and complete population-time sensitivity. The representative approach has two lanes upstream of a finite left-turn bay and three lanes at the stop line. Numerical calibration, baseline implementation checks and empirical validation are separate from this mathematical definition.

1 Traffic world, action and physical ownership

Retain one isolated four-leg junction, fixed routes and four protected effective-service groups: NS through/right, NS left, EW through/right, EW left. With physical capacity C_ℓ and saturation $s_{\ell m}$, the service law below adapts Han et al.'s average-signal interpretation to the task's four groups and capacity bound. Its connection to the source is explained in Section 5.

$$g_p \geq 0, \quad \sum_{p=1}^{4} g_p = 1, \qquad \mu_{\ell m}(g) = \min\{C_\ell, s_{\ell m} g_{p(m)}\}. \tag{1}$$

Mass is continuous vehicle-equivalent quantity; C, s, μ are veh-eq/s. A feasible direction preserves the simplex. Route labels, permissions, topology and source-order skeleton are structural. No phase waveform, integerization, microscopic lane changing or endogenous route choice is introduced. Control uses constant g during the declared control window, then a common candidate-independent clearance allocation.

The representative approach has two parallel feeder lanes and three stop-line lanes. The following graph defines the representative task geometry:

$$U \longrightarrow \{F_1, F_2\}, \qquad F_1 \longrightarrow \{B, T_1\}, \qquad F_2 \longrightarrow T_2. \tag{2}$$

U is the aggregate upstream road. Within the local zone, F_1 carries left/through traffic and F_2 through/right traffic. The internal bay opening divides F_1 into left-only bay B and through-only continuation T_1; the same cross-section divides F_2 from shared through/right continuation T_2. In the symmetric core $L_{F_1} = L_{F_2}$ and $L_B = L_{T_1} = L_{T_2}$, giving equal total approach path length. All sections own disjoint physical road space and storage.

Left traffic may enter only F_1, then B; right traffic only F_2, then T_2. Through traffic may choose F_1 or F_2 at actual admission from U, but follows its chosen lane thereafter: F_1 to T_1 or F_2 to T_2. There are no cross-feeder lane-change edges and no direct U to B, T_1, T_2 edges. The queue estimate at the choice boundary uses current receiving-feeder mass and earlier actual admissions, as in the allocation rule below; no downstream look-ahead or new driver cost is added. The two physical lanes can therefore differ in accessibility without allowing traffic to escape a lane it already occupies.

V0 uses movement-exclusive full-length paths. V1 uses a shared lane. V2/V3 share the representative bay geometry and permissions: V2 suppresses order-sensitive obstruction by construction, whereas V3 tests it.

V4 reserves demand/turn-share, bay-length/storage, arrival-order and downstream-receiving combinations, including asymmetric approaches, from calibration and method tuning. Those scenario variations test robustness after the isolated and combined mechanism ablations.

The smaller single-feeder case remains a local mechanism diagnostic. Permissions may differ between scenario families, but not between compared models or paired order cases of the same scenario.

Processing order and physical lane order. An unassigned link-end or external-source pool has an ordered processing record. A realized local lane has an actual first-in, first-out sequence. The input record governs allocation inspections; the occupied lane also imposes front blocking. A waiting head in F_1 cannot be passed by a later resident, while an independent F_2 has its own available path. A record cannot be relabeled as unassigned to permit physical overtaking.

2 Propagation and the step clock

Each physical section retains cumulative counts $N^+_{e,k}, N^-_{e,k}$ and occupancy $n_{e,k} = N^+_{e,k} - N^-_{e,k}$ at $t_k = k\Delta t$. Use triangular-FD capacity C_e, free-flow delay τ^f_e, backward delay τ^b_e and storage $K_e = C_e(\tau^f_e + \tau^b_e)$, with $\Delta t \le \min_e(\tau^f_e, \tau^b_e)$. The storage identity and timestep domain are inherited from the project's triangular-FD LTM transcription. Road capacities, delays and legal saturation rates are finite and positive. Section lengths and timestep are chosen to satisfy this domain.

Reuse that transcription's affine boundary-history reconstruction. With empty prehistory and $\theta = (t - t_h)/\Delta t$,

$$\mathcal{H}^{\pm}_e(t) = (1-\theta)N^{\pm}_{e,h} + \theta N^{\pm}_{e,h+1}, \qquad t_h \le t \le t_{h+1}. \tag{3}$$

The propagation basis is Yperman's LTM, using the repository's verified discrete transcription. The author-uploaded thesis is the literature source; the internal feeder graph is the present physical composition.

The sending count subtracts prior departures from entries early enough to reach the exit. The receiving count starts with storage, adds departures whose space-release wave has reached the entrance, and subtracts all previous entries. Both counts are bounded by capacity times timestep. The two limits below reuse that discrete LTM transcription on the task's internal section graph:

$$A_{e,k} = [\mathcal{H}^+_e(t_{k+1} - \tau^f_e) - N^-_{e,k}]_+, \qquad S_{e,k} = \min(C_e\Delta t, A_{e,k}), \tag{4}$$

$$R_{e,k} = \min\{C_e\Delta t,\ [K_e + \mathcal{H}^-_e(t_{k+1} - \tau^b_e) - N^+_{e,k}]_+\}. \tag{5}$$

The positive part sets negative availability to zero. Free-flow delay controls downstream travel; backward delay controls how soon new room is available upstream. A is ready resident mass for this update; S, R are mass allowances, not instantaneous rates. Queries use already stored boundaries. New inflow cannot traverse its receiving section in the same update.

This is a timestep model. The internal head-service calculation produces endpoint totals; it does not add propagation-history knots. Sending eligibility is evaluated for the whole update from the stored boundary histories; it does not require every selected portion to be ready at every internal service instant. The same propagation law applies to each section of the approach graph.

3 Ledger, eligibility and lane membership

Link-wise route tracking on a macroscopic backbone is a literature idea; de Souza et al. (2025) achieve individual tracking using integer flows. That reference motivates identity tracking; no integer-flow formula is imported here. The representation here instead retains continuous packet quantities with fixed route/order metadata. A long link has a route-mass ledger; each realized local lane has its own ordered ledger. The continuously changing mass and timing fields, rather than route IDs or permutation choices, carry derivatives. The exact split/merge and timing construction follows the repository's continuous route-mass specification.

Store positive-mass cohorts, removing exact zero entries without a cutoff. A cohort stores route-table reference, route position, continuous mass, current owner and affine eligibility provenance. Current movement is projected from its route. Internal $U/F_1/F_2/B/T_1/T_2$ crossings preserve route position; the actual junction crossing advances it. Future movement sequences are not duplicated in grid entries.

Only actual admission establishes receiving-lane membership. Unadmitted mass stays with its old owner without a binding future-lane vector. Traffic already in F_1 or F_2 retains that realized lane; its movement and the declared connectivity determine its continuation. Resident traffic cannot be reassigned across feeders or terminal lanes to improve balance.

Retain the exact source order of the unserved record. Before evaluating an operator, coalesce exactly compatible adjacent representation fragments: same route/position, owner and permissions, with contiguous

eligibility segments lying on the same affine timing line. Exact merging preserves mass, route identity, FIFO order and timing; no tolerance-based route mixing is used.

The near-intersection movement grid and the lane ledger represent the same mass in complementary forms. The grid supplies per-lane continuous movement amounts and loads; the ledger resolves the exact order hidden by a mixed bin. Classical CTM (Daganzo, 1994) uses physical-space cells. Here fixed cells are cumulative-mass/rank intervals: a mass-coordinate adaptation, not an additional spatial CTM propagation law. Physical capacities, travel and storage remain governed by the section LTM histories and geometry.

Use a fixed equal-mass lattice with geometry-derived capacity masks, including a possible partial final bin. For packet r on lane ℓ, let $I_{\ell r} = [z^-_{\ell r}, z^+_{\ell r}]$, where $z^-_{\ell r} = \sum_{v<r} q_{\ell v}$ and $z^+_{\ell r} = z^-_{\ell r} + q_{\ell r}$. Bin c has fixed boundaries $\zeta_{\ell c}, \zeta_{\ell,c+1}$. The overlap formula is derived from interval intersections in the project's fixed cumulative-mass grid; CTM supplies cell-based inspiration, not this flux-free representation:

$$\begin{aligned} \omega_{\ell cr} &= [\min(z^+_{\ell r}, \zeta_{\ell,c+1}) - \max(z^-_{\ell r}, \zeta_{\ell c})]_+, \\ M_{\ell cm} &= \sum_{r:m_r=m} \omega_{\ell cr}, \qquad n_\ell = \sum_{c,m} M_{\ell cm} = \sum_r q_{\ell r}. \end{aligned} \tag{6}$$

Here $\omega_{\ell cr}$ is packet r's overlap mass in bin c, $M_{\ell cm}$ is movement m's mass in that lane/bin, and n_ℓ is total resident lane mass, including moving and waiting traffic. Their units are vehicle-equivalents, not metres. This formula reconstructs the grid from the ledger; it is not Daganzo's cell flux equation. Rebuild it atomically after conservative service and admission, keeping zero cells in fixed padded shapes. Changing the mass resolution changes neither storage nor geometry. The grid is a derived continuous state, not another stock to add to the mass account. A mixed bin cannot reconstruct the head sequence, so exact lane access/service continues to use the ledger.

Sending limits transfers, not candidate inspections. Form candidates from ready mass A, cutting an eligibility segment exactly if needed. For an unassigned pool, scan these candidates and spend at most S in total. The candidate list is not restricted to the first S units before access is checked: an earlier blocked movement must not hide later ready traffic with an independently accessible path. For a physical lane, the access rule below forces actual served mass to be a prefix.

This adapts the continuous-ledger ready-prefix construction by separating candidate inspection from the total transferred allowance. After a selection that skips an independently blocked movement, unserved mass keeps its original entry/travel history and relative residual order. Waiting for admission does not restart its travel delay. Every past departure was eligible when served, and eligibility is monotone. Total eligible residual mass therefore remains $\mathcal{H}^+(t_{k+1} - \tau^f) - N^-_k$ even if eligible departures were not a global processing-order prefix. An occupied lane still serves its physical FIFO prefix.

4 Allocation, unassigned residuals and physical access

The equilibrium allocation idea is motivated by Gong, Bliemer and Raadsen (2025). The raw queue-mass cost and capped per-packet entry rule below are the present adaptation. They are derived from a strictly convex load-balancing problem, not claimed as that paper's original exit-resistance equation or general algorithm.

Let a be the sending owner, r a ready cohort of mass q_r and movement m_r, and ℓ a receiving lane/section. Permission $P_{m\ell} \in \{0,1\}$ and reachability $E_{r\ell} \in \{0,1\}$ encode actual paths/access, not a future green plan.

Initialize residual sending $s = S_{a,k}$, residual receiving $R^{\text{rem}}_\ell = R_{\ell,k}$ and perceived loads $h_\ell = n_{\ell,k}$. Process cohorts in recorded order. Let $\mathcal{B}$ contain movements already left partly unserved in this scan. Each sending owner has its own scan and blocked-movement set; one inbound leg does not block another. If $m_r \in \mathcal{B}$ set $V_r = x_{r\ell} = 0$; otherwise admit the largest feasible total V_r, limited by packet mass, remaining sending and legal reachable receiving. Distribute that total to reduce differences in current lane loads, using a common filling level λ_r.

The quadratic objective and its capped solution below are defined and derived for this entry problem; Gong et al. supply the equilibrium motivation.

$$u_{r\ell} = P_{m_r \ell} E_{r\ell} R_\ell^{\text{rem}}, \qquad V_r = \min\{q_r, s, \sum_\ell u_{r\ell}\},$$

$$x_r = \operatorname*{arg\,min}_{0 \le x_{r\ell} \le u_{r\ell},\ \sum_\ell x_{r\ell} = V_r} \frac{1}{2} \sum_{\ell : P_{m_r \ell} = 1} (h_\ell + x_{r\ell})^2,$$

$$x_{r\ell} = \min\{u_{r\ell}, [\lambda_r - h_\ell]_+\}, \qquad \sum_\ell x_{r\ell} = V_r. \tag{7}$$

The solver chooses λ_r so the lane additions sum to V_r. Its units are vehicle-equivalent mass: $\lambda_r - h_\ell$ is the mass needed to bring a lane to that load. The positive part excludes lanes already above the level and $u_{r\ell}$ caps each feasible admission. This is a mass-balancing rule, not a stop-line service cap. Illegal lanes have zero addition. Strict convexity selects unique additions; the filling level itself need not be unique when all usable lanes are capped. For $V = 0$, use $x = 0$. Commit only actual entry. The following resource updates are conservation identities derived from the admitted amounts in (7):

$$s \leftarrow s - V_r, \quad R_\ell^{\text{rem}} \leftarrow R_\ell^{\text{rem}} - x_{r\ell}, \quad h_\ell \leftarrow h_\ell + x_{r\ell}, \quad q_r^{\text{rem}} = q_r - V_r. \tag{8}$$

If $q_r^{\text{rem}} > 0$, retain it at its original position and add m_r to $\mathcal{B}$ for this update. Later cohorts of that movement wait; other movements are considered. Reset $\mathcal{B}$ next update. The retained record then puts the older residual first within its movement.

The perceived load h is pre-state local lane mass plus earlier actual admissions in the current scan. It uses current mass without forecasting later signal discharge. The physical occupancy update separately accounts for that step's exits.

Access in the admitted layouts

For an unassigned pool with independently accessible receiving lanes, $E_{r\ell} = 1$ on those paths. Movement labels alone do not create access. This applies at $U \to \{F_1, F_2\}$ on a packet's legal feeder choices, and to a source feeding independent V0 paths. Once mass occupies F_1 or F_2, it cannot use the other feeder's capacity.

For an occupied single lane, including F_1 at its fork, the access predicate is defined from realized-lane FIFO and the chosen connectivity:

$$E_{r\ell} = \mathbf{1}\{\text{every earlier resident cohort has fully crossed}\} \quad \text{on a legal outgoing edge.} \tag{9}$$

An older physical residual prevents later physical entry even while the allocator inspects later records. Several actual lanes have separate ledgers/access conditions, not one global FIFO.

Movement-specific blocking governs the unassigned scan; actual lane occupation adds the physical access constraint. At U, a full F_1 prevents left entry while through traffic may still use reachable F_2. At the fork, a through head in F_1 can obstruct a later turner despite an empty bay. A full bay can similarly leave a left head blocking later through traffic in F_1. Independently resident F_2 traffic may continue toward T_2 if its own sending, receiving and order permit it.

A zero left-turn stop-line rate is not automatically zero lane-entry room. A left turner may join a reachable feeder or bay with room and subsequently contribute to spillback. In this geometry it cannot turn left from T_1 or use F_2, T_2; a full bay has no alternative left-turn stop-line lane. Each constraint refers to its actual physical boundary.

Direct definition checks

The following local reference cases distinguish an unassigned pool with independent access from an occupied feeder with physical FIFO. For an independent-access pool with order $[L(0.4), T(0.6)]$, $A = 1$, $S = 0.6$ and exclusive-destination receiving $(0, 0.5)$, left entry is zero and through entry 0.5. Residual order is $[L(0.4), T(0.1)]$. Truncating candidates to the first S mass would expose only 0.2 through mass and give the wrong result for this pool rule.

For actual F_1 take $[T(0.4), L(0.6)]$, $S_{F_1} = 1$, $R_{T_1} = 0.2$, $R_B = 1$. Through entry is 0.2, left entry zero. Reverse the order with the same totals and branch loads $(n_B, n_{T_1}) = (0, 1.8)$. Left uses B first, then

through enters 0.2: total 0.8 rather than 0.2. These branch bounds are compatible with $\Delta t = \tau^f = \tau^b = 1$, $C_B = C_{T_1} = 1$, $K_B = K_{T_1} = 2$. If an independent ready 0.3 in F_2 has enough receiving in T_2, it enters in either case. Cross-section totals are then 0.5 and 1.1. These are hand-calculated operator reference cases. Full task scenarios use loading from empty.

At the U choice boundary, a compatible balancing case has feeder loads $(1.2, 1.0)$, receiving $(0.8, 1.0)$, ready through mass 1 and sending 0.6. Additions $(0.2, 0.4)$ leave perceived loads $(1.4, 1.4)$; the remaining upstream 0.4 is unassigned.

5 Elapsed service and selected receiving rates

The use of green fractions as average service follows the continuum signal interpretation of Han et al. (2014). The conservation/supply framework follows first-order node modeling; Tampere et al. (2011) is a basis for those requirements. The particular green-weighted leg sharing and equal-within-leg capped reuse below are model-specific priorities, rather than a copy of the reference's oriented-capacity rule.

A stop-line lane exposes its current ready resident head, with sending quota S_ℓ and cap $c_\ell = \mu_{\ell m}(g)$. The constant-rate conversion below is the project's chosen average-receiving convention, applied to the LTM allowance:

$$\rho_j = R_{j,k}/\Delta t. \tag{10}$$

This is a timestep computational average, not an inferred within-step physical supply curve.

For each output j, collect current heads by inbound leg i. A leg-to-output movement here has one service group. The next two equations define this model's green-weighted leg and equal-lane priority rules within the cited feasibility framework:

$$C_{ij} = \sum_{\ell \in \mathcal{L}_{ij}} c_\ell, \quad w_{ij} = g_{p(i,j)}, \quad W_j = \min(\rho_j, \sum_i C_{ij}),$$

$$r_{ij} = \min(C_{ij}, \theta_j w_{ij}), \quad \sum_i r_{ij} = W_j, \tag{11}$$

$$r_\ell = \min(c_\ell, \eta_{ij}), \quad \sum_{\ell \in \mathcal{L}_{ij}} r_\ell = r_{ij}. \tag{12}$$

Remove zero-cap competitors; zero W gives zero rates.

Here θ_j is chosen so green-weighted leg rates total W_j; η_{ij} is the common uncapped lane rate chosen so the lane rates total r_{ij}. Its units are vehicle-equivalents/s, unlike the mass-level λ_r used for entry. Each lane receives its cap or that common rate, whichever is smaller. Thus a grant 0.6 splits 0.3/0.3 between two unrestricted lanes; with one capped at 0.1, the other may use 0.5. The leg weight is counted once, independent of lane count.

Rates describe the current heads; their one-step amounts require the time needed to clear each head. For a one-second shared-lane example, a ready front block of mass 0.2 at rate 0.4 uses half a second, leaving half a second for an already-ready following movement at rate 0.6, which then serves 0.3. With several lanes, a changed head can alter receiving competition. Let σ be time already accounted for, starting at zero, and q_ℓ the residual head mass. Current rates apply until a head clears, remaining sending is spent or the step ends:

Combining the current-head, sending and timestep stopping constraints derives the duration rule:

$$\delta = \min\left\{\Delta t - \sigma, \min_{\ell : r_\ell > 0} \frac{\min(q_\ell, S_\ell^{\text{rem}})}{r_\ell}\right\}. \tag{13}$$

Transfer $r_\ell \delta$, process tied events together, expose new heads and recompute sharing. Stop when rates vanish or time ends. For a chosen lane/movement stream, r_a is its rate in interval a, zero when it is not the current head. Summing the transferred mass rate times duration derives its recorded timestep amount:

$$y = \sum_a r_a \delta_a. \tag{14}$$

Include remaining idle time with zero rate when describing the full timestep. This accounting does not add propagation-history knots. A zero-rate head blocks its lane. Newly admitted mass is outside this resident service tranche.

A later ready head joins current competition; earlier discharge is not compensated to equalize final totals. Constant receiving removes a separate consumable-pool exhaustion rule, but head clearance and changing competitors remain sequential. Receiving reuse does not transfer one signal group's service to another.

For V1's separate shared left/through service diagnostic, ready left mass q_L precedes sufficient through mass. With nonbinding receiving and physical caps, its clearance time determines the remaining through service time. The example below is derived by applying (14) to a head-clearing left block and its through follower:

$$y_L = q_L, \qquad y_T = s_T g_T \left(\Delta t - \frac{q_L}{s_L g_L} \right), \quad 0 < \frac{q_L}{s_L g_L} < \Delta t. \tag{15}$$

With unit timestep/saturations, $q_L = 0.2$ and $(g_L, g_T) = (0.4, 0.6)$, through discharge is 0.3. This is the optional algebraic check of the rate-versus-time mechanism explained in the main report. For $g_L(\alpha) = g_L + \alpha$ and $g_T(\alpha) = g_T - \alpha$, differentiating (15) by the product and quotient rules gives

$$\frac{\mathrm{d}y_T}{\mathrm{d}\alpha} = -s_T \left(\Delta t - \frac{q_L}{s_L g_L} \right) + \frac{s_T g_T q_L}{s_L g_L^2}. \tag{16}$$

At the stated values the rate-loss term is -0.5 and the earlier-clearance term is 0.75, giving 0.25. These are the two channels of (26); (27) supplies the head-time response. A fully cleared fixed head contributes no extra mass itself; its followers can use the time saved. This is not a complete-TST benefit or an aggregate-rival failure claim.

6 Paired transfers, histories and feasibility

A one-in/one-out internal edge transfers its ready prefix up to $\min(S_{\mathrm{up}}, R_{\mathrm{down}})$. Forks use allocation/access; the central junction uses elapsed service. An external source adds $D_{o,k}$ to its waiting record before admission, with budget $Q_{o,k} + D_{o,k}$. A sink removes at most sending and its declared allowance.

Each internal entry interface has one sending owner; U has two receiving feeder choices sharing its total sending allowance. Shared receiving at the central signal node uses the simultaneous current-head sharing law, rather than an arbitrary serial order of inbound legs. All nodes read pre-state histories/ledgers. Each section's sending and receiving is spent once at its boundary. New admissions are appended after selecting resident service; today's exits create no immediate receiving credit. The cumulative-count update below reuses the project's conservative LTM/ledger bookkeeping:

$$N^+_{e,k+1} = N^+_{e,k} + b_{e,k}, \quad N^-_{e,k+1} = N^-_{e,k} + y_{e,k}, \quad n_{e,k+1} = n_{e,k} + b_{e,k} - y_{e,k}. \tag{17}$$

Internal flow leaves one owner and enters the next with identical mass and route identity. Source admission removes source waiting; sink exit removes road mass. No residual is duplicated as lane stock.

The next update needs to distinguish traffic that has reached service from mass still travelling. Each admitted packet therefore carries an entry interval consistent with the uniformly reconstructed cumulative history. In a receiving section with total entrance $b > 0$, a child of mass q^{in} preceded by admitted mass z receives the interval below. Its endpoints are derived from the uniform boundary reconstruction inherited from the continuous ledger specification:

$$[t^{\mathrm{in}}_{\mathrm{front}}, t^{\mathrm{in}}_{\mathrm{tail}}] = \left[t_k + \Delta t \, \frac{z}{b}, \; t_k + \Delta t \, \frac{z + q^{\mathrm{in}}}{b} \right]. \tag{18}$$

Both endpoints are entry times: the first and last parts of that child. For $b = 1$, $z = 0.2$, $q^{\mathrm{in}} = 0.3$ and unit timestep, the child enters from 0.2 to 0.5 seconds after the update starts. Add section free-flow travel time τ^f_e to both endpoints to obtain its earliest exit readiness interval. Retained portions keep that timing; waiting does not restart travel. A later timestep can select only the ready fraction. Zero inflow has no admitted child and therefore no entry interval.

Children are appended in actual admission order. For cross-input outgoing bookkeeping, order by service-interval start, then fixed input leg, lane and source-cohort indices. That serialization does not change shared-rate grants. The outgoing roads lead to sinks, so it cannot create a later signal-order advantage in this task.

The timestep domain gives $S_{e,k} \le n_{e,k}$ and $R_{e,k} \le K_e - n_{e,k}$. Combining these inherited bounds with (17) derives the storage implication below:

$$0 \le b_{e,k} \le R_{e,k}, \quad 0 \le y_{e,k} \le S_{e,k} \implies 0 \le n_{e,k} + b_{e,k} - y_{e,k} \le K_e. \tag{19}$$

Summing the paired conservation updates from empty loading, including source waiting and sink exits, derives

$$\sum_{o} Q_{o,k} + \sum_{e} n_{e,k} + Z_k = D_{<k}, \tag{20}$$

also route by route. Z_k is cumulative sink exit; $D_{<k}$ all earlier exogenous release.

A finite state has a defined unique result: finite source scan, unique convex allocation vectors, unique capped-sharing vectors and a finite head-event calculation. Each positive service interval exhausts a head/quota or reaches step end. Exact ties are handled together; no mass cutoff or software priority manufactures uniqueness. These are definition-level arguments, not implementation or traffic-fidelity evidence.

7 Sensitivities through the traffic modules

Let $g(\alpha) = g + \alpha d$, with $\sum_p d_p = 0$, define a feasible service transfer. All derivatives below are with respect to that same control amount α. Fix routes, permissions and the current active head/constraint pattern. A structural switch requires its own directional branch.

The path is service fractions to head rates, then actual transfers, then continuous ledger masses/timing, grid overlaps and road histories, then later sending/receiving and admission, and finally sink exits and complete time. The grid is reconstructed inside the step map; labels and order are structural while masses and timing are differentiated.

Differentiate the admission rule and its mass balance in (7). For the current packet, let $\mathcal{A}$ contain positive uncapped lanes ($0 < x_\ell < u_\ell$) and $\mathcal{U}$ lanes on binding upper bounds. Recall V is total admitted mass, h_ℓ the load before this allocation, u_ℓ its entry bound and λ the common filling level. The balance and cases below are derivatives of (7) on fixed active categories:

$$\begin{aligned} \frac{\mathrm{d}V}{\mathrm{d}\alpha} &= \sum_{\ell} \frac{\mathrm{d}x_\ell}{\mathrm{d}\alpha}, \\ \frac{\mathrm{d}x_\ell}{\mathrm{d}\alpha} &= \begin{cases} \dfrac{\mathrm{d}\lambda}{\mathrm{d}\alpha} - \dfrac{\mathrm{d}h_\ell}{\mathrm{d}\alpha}, & \ell \in \mathcal{A}, \\ \dfrac{\mathrm{d}u_\ell}{\mathrm{d}\alpha}, & \ell \in \mathcal{U}, \\ 0, & \text{no addition on the unchanged lower branch.} \end{cases} \end{aligned} \tag{21}$$

Substituting these lane responses into the total and solving for the common filling-level sensitivity in (21) derives the algebraic solution

$$\frac{\mathrm{d}\lambda}{\mathrm{d}\alpha} = \frac{1}{|\mathcal{A}|} \left(\frac{\mathrm{d}V}{\mathrm{d}\alpha} - \sum_{\ell \in \mathcal{U}} \frac{\mathrm{d}u_\ell}{\mathrm{d}\alpha} + \sum_{\ell \in \mathcal{A}} \frac{\mathrm{d}h_\ell}{\mathrm{d}\alpha} \right). \tag{22}$$

The capped-bound response consumes part of the admitted-mass response; starting-load increases require a higher level to preserve free additions. The denominator counts positive uncapped receivers. Evaluate this level response first, then the individual additions. This carries prior traffic changes into actual later lane entry, not a new direct dependence on green.

Stable fully capped branches follow their caps; stable zero admission has zero admission derivative. Differentiate V through its limiting argument and u through the current receiving/access branch. Sequential updates differentiate the resource updates in (8) and preserve the response of remaining mass and resources. The following is direct differentiation of those resource identities, not a new allocation rule:

$$\begin{aligned} \frac{\mathrm{d}h_\ell}{\mathrm{d}\alpha} &\leftarrow \frac{\mathrm{d}h_\ell}{\mathrm{d}\alpha} + \frac{\mathrm{d}x_\ell}{\mathrm{d}\alpha}, \\ \frac{\mathrm{d}R_\ell^{\mathrm{rem}}}{\mathrm{d}\alpha} &\leftarrow \frac{\mathrm{d}R_\ell^{\mathrm{rem}}}{\mathrm{d}\alpha} - \frac{\mathrm{d}x_\ell}{\mathrm{d}\alpha}, \\ \frac{\mathrm{d}s}{\mathrm{d}\alpha} &\leftarrow \frac{\mathrm{d}s}{\mathrm{d}\alpha} - \frac{\mathrm{d}V}{\mathrm{d}\alpha}, \qquad \frac{\mathrm{d}q^{\mathrm{rem}}}{\mathrm{d}\alpha} = \frac{\mathrm{d}q}{\mathrm{d}\alpha} - \frac{\mathrm{d}V}{\mathrm{d}\alpha}. \end{aligned} \tag{23}$$

Green directly changes the signal-limited rate in (1), and changes the weights in (11); receiving changes through (10). Recall r_{ij} is the leg grant, C_{ij} its cap, w_{ij} its weight, W_j the usable total and θ_j the common

multiplier. For uncapped legs $\mathcal{F}$ and capped legs $\mathcal{C}$, differentiating the model-defined (11) on fixed cap categories derives

$$\begin{aligned}\frac{\mathrm{d}W_j}{\mathrm{d}\alpha} &= \sum_i \frac{\mathrm{d}r_{ij}}{\mathrm{d}\alpha},\\ \frac{\mathrm{d}r_{ij}}{\mathrm{d}\alpha} &= \begin{cases} \dfrac{\mathrm{d}C_{ij}}{\mathrm{d}\alpha}, & i \in \mathcal{C},\\ w_{ij}\dfrac{\mathrm{d}\theta_j}{\mathrm{d}\alpha} + \theta_j \dfrac{\mathrm{d}w_{ij}}{\mathrm{d}\alpha}, & i \in \mathcal{F}. \end{cases}\end{aligned} \tag{24}$$

Substitute these responses into the total-rate balance, then solve for the common multiplier. This algebraic rearrangement of (24) gives

$$\frac{\mathrm{d}\theta_j}{\mathrm{d}\alpha} = \frac{\dfrac{\mathrm{d}W_j}{\mathrm{d}\alpha} - \displaystyle\sum_{i\in\mathcal{C}} \frac{\mathrm{d}C_{ij}}{\mathrm{d}\alpha} - \theta_j \sum_{i\in\mathcal{F}} \frac{\mathrm{d}w_{ij}}{\mathrm{d}\alpha}}{\displaystyle\sum_{i\in\mathcal{F}} w_{ij}}. \tag{25}$$

Subtract the capped grants and the response already supplied by changing weights; normalize the remaining response by the free legs' total weight. Evaluate the multiplier response, then each leg's response. If no leg is uncapped, differentiate its caps directly; stable zero supply gives zero rates. Differentiate (12) by the same balance with equal lane weights. This supplies the current-rate channel of discharge.

Differentiate the service-amount sum (14) to turn rate and duration changes into an actual transferred-mass response. The next equation is its product-rule derivative:

$$\frac{\mathrm{d}y}{\mathrm{d}\alpha} = \sum_a \left(\delta_a \frac{\mathrm{d}r_a}{\mathrm{d}\alpha} + r_a \frac{\mathrm{d}\delta_a}{\mathrm{d}\alpha} \right). \tag{26}$$

The first term changes rates over current durations; the second changes time available at current rates.

Equation (13) selects the duration branch. With the same controlling lane ℓ and stopping condition, differentiating that selected branch derives the three cases below:

$$\frac{\mathrm{d}\delta_a}{\mathrm{d}\alpha} = \begin{cases} \dfrac{1}{r_{\ell,a}}\dfrac{\mathrm{d}q_{\ell,a}}{\mathrm{d}\alpha} - \dfrac{q_{\ell,a}}{r_{\ell,a}^2}\dfrac{\mathrm{d}r_{\ell,a}}{\mathrm{d}\alpha}, & \text{head clearance,}\\ \dfrac{1}{r_{\ell,a}}\dfrac{\mathrm{d}S^{\mathrm{rem}}_{\ell,a}}{\mathrm{d}\alpha} - \dfrac{S^{\mathrm{rem}}_{\ell,a}}{r_{\ell,a}^2}\dfrac{\mathrm{d}r_{\ell,a}}{\mathrm{d}\alpha}, & \text{sending allowance spent,}\\ -\displaystyle\sum_{b<a}\frac{\mathrm{d}\delta_b}{\mathrm{d}\alpha}, & \text{remaining timestep time.} \end{cases} \tag{27}$$

Head mass and remaining allowance here already include preceding intervals' transfers. The first branch differentiates $r\delta = q$: extra mass needs time, whereas faster service clears the head earlier. It is the head-clearance path emphasized in the main report, not the derivative of all minima at once. In the final branch, $\sigma = \sum_{b<a}\delta_b$ and Δt is fixed. Include any idle remainder with zero rates. Tied stopping conditions retain their directional/active-pattern treatment; the formula asserts no unique ordinary derivative at a switch. These duration responses supply the time channel in (26).

After service and admission, packet masses determine their cumulative rank endpoints. Fixed bin boundaries give the following continuous path through direct differentiation of the project representation (6):

$$\begin{aligned}\frac{\mathrm{d}z^-_{\ell r}}{\mathrm{d}\alpha} &= \sum_{v<r}\frac{\mathrm{d}q_{\ell v}}{\mathrm{d}\alpha}, \qquad \frac{\mathrm{d}z^+_{\ell r}}{\mathrm{d}\alpha} = \frac{\mathrm{d}z^-_{\ell r}}{\mathrm{d}\alpha} + \frac{\mathrm{d}q_{\ell r}}{\mathrm{d}\alpha},\\ \frac{\mathrm{d}M_{\ell cm}}{\mathrm{d}\alpha} &= \sum_{r:m_r=m}\frac{\mathrm{d}\omega_{\ell cr}}{\mathrm{d}\alpha}, \qquad \frac{\mathrm{d}n_\ell}{\mathrm{d}\alpha} = \sum_{c,m}\frac{\mathrm{d}M_{\ell cm}}{\mathrm{d}\alpha}.\end{aligned} \tag{28}$$

Differentiate each overlap through its active min/max endpoints. The cell boundary is fixed; the packet boundary moves. Fixed metadata are not required to carry gradients. Exact-compatible merging preserves the same overlap and timing map.

For a positive-width readiness interval $[t_{\mathrm{f}}, t_{\mathrm{t}}]$ and fixed cutoff time t, uniform readiness timing derives the interior fraction below; quotient and product rules then derive its sensitivities. The first line defines that fraction:

$$\begin{aligned} f &= \frac{t - t_{\mathrm{f}}}{t_{\mathrm{t}} - t_{\mathrm{f}}}, \\ \frac{\mathrm{d}f}{\mathrm{d}\alpha} &= \frac{-\dfrac{\mathrm{d}t_{\mathrm{f}}}{\mathrm{d}\alpha} - f\left(\dfrac{\mathrm{d}t_{\mathrm{t}}}{\mathrm{d}\alpha} - \dfrac{\mathrm{d}t_{\mathrm{f}}}{\mathrm{d}\alpha}\right)}{t_{\mathrm{t}} - t_{\mathrm{f}}}, \\ \frac{\mathrm{d}q^{\mathrm{ready}}}{\mathrm{d}\alpha} &= f\frac{\mathrm{d}q}{\mathrm{d}\alpha} + q\frac{\mathrm{d}f}{\mathrm{d}\alpha}. \end{aligned} \tag{29}$$

Use whole/empty branches outside the interval. Point segments use their boundary branch. For positive total entrance b, differentiate the front-entry endpoint of (18) by the quotient rule to obtain

$$\frac{\mathrm{d}t^{\mathrm{in}}_{\mathrm{front}}}{\mathrm{d}\alpha} = \Delta t\left(\frac{1}{b}\frac{\mathrm{d}z}{\mathrm{d}\alpha} - \frac{z}{b^2}\frac{\mathrm{d}b}{\mathrm{d}\alpha}\right). \tag{30}$$

The tail replaces z by $z + q^{\mathrm{in}}$. Zero entrance creates no segment. The free-flow time is fixed. Delayed cumulative queries keep fixed interpolation weights but differentiate stored boundary counts. On positive eligible-mass and storage-limited receiving branches, direct differentiation of the inherited (4) and (5) gives:

$$\begin{aligned} \frac{\mathrm{d}A_{e,k}}{\mathrm{d}\alpha} &= \frac{\mathrm{d}\mathcal{H}^+_e(t_{k+1} - \tau^f_e)}{\mathrm{d}\alpha} - \frac{\mathrm{d}N^-_{e,k}}{\mathrm{d}\alpha}, \\ \frac{\mathrm{d}R_{e,k}}{\mathrm{d}\alpha} &= \frac{\mathrm{d}\mathcal{H}^-_e(t_{k+1} - \tau^b_e)}{\mathrm{d}\alpha} - \frac{\mathrm{d}N^+_{e,k}}{\mathrm{d}\alpha}. \end{aligned} \tag{31}$$

Travel-limited sending follows $\mathrm{d}A/\mathrm{d}\alpha$; a fixed capacity-limited sending or receiving branch has zero derivative. Stable zero branches give zero availability response. These branch rules are the derivatives of the existing LTM limits, not a new propagation law.

8 Complete population time and continued state response

Collect source masses, cumulative histories and continuous ledger mass/timing fields in X_k. Per-lane grids are deterministic functions of those data inside F_k. If g_k denotes applied service allocation, use $X_{k+1} = F_k(X_k, g_k)$ and the complete population account below. The update map denotes the present model composition; the objective reuses the LTM project's source-plus-road time quadrature:

$$J_{\Delta t} = \Delta t \sum_{k=0}^{H-1}\left[\sum_o (Q_{o,k} + D_{o,k}) + \sum_e n_{e,k}\right]. \tag{32}$$

Q is existing source waiting, D newly available release, n all road mass and H the common complete-clearance horizon. The time charge has vehicle-equivalent-second units. Source admission transfers ownership but does not remove population from the account.

Let c_k be the timestep charge in (32). Differentiating the displayed state update $X_{k+1} = F_k(X_k, g_k)$ and summing these charges derives the following chain-rule identities from those state and objective definitions:

$$\begin{aligned} \frac{\mathrm{d}X_{k+1}}{\mathrm{d}\alpha} &= \frac{\partial F_k}{\partial X_k}\frac{\mathrm{d}X_k}{\mathrm{d}\alpha} + \frac{\partial F_k}{\partial g_k}\frac{\mathrm{d}g_k}{\mathrm{d}\alpha}, \\ \frac{\mathrm{d}J}{\mathrm{d}\alpha} &= \sum_k\left(\frac{\partial c_k}{\partial X_k}\frac{\mathrm{d}X_k}{\mathrm{d}\alpha} + \frac{\partial c_k}{\partial g_k}\frac{\mathrm{d}g_k}{\mathrm{d}\alpha}\right). \end{aligned} \tag{33}$$

During control $\mathrm{d}g_k/\mathrm{d}\alpha = d$; during fixed common clearance it is zero, while the state response persists. Candidate-independent empty loading/preparation gives initial response zero. At a fixed state the present population charge has zero direct green partial. Stop-line service responds directly; its later queue/history changes alter admission, spillback, source waiting and eventual exit.

For fixed releases, let $D_{\le k}$ be cumulative available release and Z_k cumulative sink exits before update k. The conservation identity (20), after adding update k's new release, equates source-plus-road population in (32) to $D_{\le k} - Z_k$. With releases, timestep and common horizon fixed, the following objective expression and derivative are algebraic consequences of (20) and (32):

$$J_{\Delta t} = \Delta t \sum_{k=0}^{H-1} (D_{\le k} - Z_k), \qquad \frac{\mathrm{d}J_{\Delta t}}{\mathrm{d}\alpha} = -\Delta t \sum_{k=0}^{H-1} \frac{\mathrm{d}Z_k}{\mathrm{d}\alpha}. \tag{34}$$

A common integer horizon is chosen to clear the compared action neighborhood, followed by zero occupancy. Finite demand, acyclic routes, positive continuation service and eventual sink availability support this choice; clearance is checked for each comparison. A nonclearing run provides a truncated population-time total and is identified as such. The derivative domain includes clearance and unchanged active signatures; switches require directional branches.

9 Matched comparison models

Models share population, geometry/permissions, action, objective, preparation, continuation and a declared data protocol. Original fine-grained series may pass through each model's own macro converter/estimator. Declare retained variables, memory, update schedule and calibration data. Native subsets are legitimate. Successful macro inference counts against needing exact order; an exact ordered ledger retained inside an estimator remains meso information.

Equal current totals do not establish equality of all lawful histories. State-conditioned replay, observation-updated prediction and forecasting from a common prefix are distinct tasks. State which observations are fixed or regenerated for each derivative claim. A later observation cannot count as a response predicted before it was observed.

The comparison models are aggregate link, structured movement-queue, lane/storage aggregate and selective ordered-mass descriptions. They explain the same physical ablations with different retained state. Queue/storage explanations are expected to suffice when their mechanisms are the only active effects; combined order/access cases test the extra value of realized sequence.

The movement-queue comparison uses Wei et al.'s link queue model, with moving/queued state and native effective green. Its full causal short-region/link/node law must be instantiated; the fixed-speed shortcut is not assumed valid in every spillback regime.

The lane/storage comparison uses Gong et al.'s lane-exit allocation with a sourced storage adaptation, for which Liu–Chang (2011) is a candidate. Exit allocation is not finite-bay entry, and native theorems do not automatically qualify that composition. Exact implementations and adapters still require physical and task-specific checks; the cited native model is not automatically a qualified baseline.

A V2 comparator failure must be understood before attributing a V3 advantage to retained order.

10 Definition checks and external usefulness

Local reference checks cover pool/feeder access, partial admission, finite storage, current-head sharing/reuse, route conservation and the source-counted objective. V0 neutral cases need matched pure service classes and nonbinding or equivalent receiving; one through/right green does not make differently constrained exits identical. In that neutral domain the laws reduce to the same sending/service/receiving minimum. Exact equivalence concerns a matched aggregate backbone; a source-native propagation difference must be qualified and cannot be credited to retained order.

Rotating the symmetric scenario and green groups must rotate results. Equal lane totals do not justify one larger FIFO lane without corresponding order, readiness, access, service and receiving. Hand/reference checks establish definition correctness, not realism.

For the richer-reference comparison, use one fixed executable mapping. With fixed cycle C^{cyc}, lost intervals totaling $L < C^{\mathrm{cyc}}$, order and offset, the conversion below is derived from the task's fixed usable-time budget and green-fraction definition:

$$G_p = (C^{\mathrm{cyc}} - L) g_p, \qquad s^{\mathrm{eff}}_{\ell m} = s^{\mathrm{phys}}_{\ell m} (C^{\mathrm{cyc}} - L)/C^{\mathrm{cyc}}. \tag{35}$$

Program admissibility and the common fixed-clearance continuation are specified with the mapping. Adding an extra minimum green would change the represented control and is excluded from this conversion. This matches nominal average opportunity, not microscopic startup/access or every phase effect. Official SUMO signal definitions support the interface, not empirical equivalence. Those definitions describe how to encode the program; they are not the source of the budget identities above.

Use a finite master route population and common desired releases. Integer reference vehicles are exogenous population, not rounded internal mass. Reference lane behavior is fixed independently of candidate directions; a candidate's private lane assignments are not imposed selectively. Reference seeds and held-out family IDs are matched, with each order pair kept together.

Reference time is desired-release-to-arrival time: `duration` plus `departDelay`, from official TripInfo. Waiting is already inside duration. Population accounting includes uninserted and unfinished trips. All demand is cleared, or noncompletion is reported; the objective is not restricted to completed trips.

For each feasible direction, compare predicted sign/ranking and scale with actual richer-evaluator complete-time change under the same action mapping. Disclose observation conditioning. Only after forward/reference and derivative checks pass should matched optimization and held-out usefulness be tested.

11 Scope and evidence limits

The model specifies eligible packet inspection, movement-specific waiting priority, actual-admission balancing, physical lane access, elapsed service, interpolated histories and complete population-time sensitivity. Its parallel approach lanes retain distinct occupancy, storage and order. The bay feeder can obstruct one part of the approach while another lane remains usable, without giving resident traffic an unmodeled bypass.

Feasibility and unique-selection arguments concern the mathematical operators. The analytic examples illustrate local consequences. Implementation correctness, numerical derivative agreement, qualified baselines and external control usefulness require separate evidence.